

A Novel Single-Cell Multi-Modal Data Integration Method Combining Similarity Network Fusion and Deep Graph Learning

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Abstract

Accurate identification of cell types is essential for studying cellular heterogeneity. Single-cell sequencing technologies measure multiple omics data types at the single-cell level, posing data integration and clustering challenges. This paper introduces a novel approach combining similarity network fusion with graph neural networks to integrate multimodal data effectively. By iteratively calculating and integrating single-cell data from different modalities, our method addresses limitations in graph neural networks graph construction for single-cell multimodal data, resulting in significant performance improvements in downstream tasks such as clustering. Our method preserves local and global cellular information and mitigates batch effects. Experimental results show that our similarity network fusion-enhanced graph neural network models outperform other methods in clustering accuracy, demonstrating improved robustness and performance for single-cell multimodal data integration.

Keywords

single-cell multimodal integration; similar fusion networks; cell clustering

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1 Introduction

Integrating multi-modal data in single-cell studies has become essential for advancing our understanding of cellular heterogeneity and complex biological systems. Multi-modal data, such as scRNA-seq (gene expression), scATAC-seq (chromatin accessibility), and CITE-seq (protein abundance), offer complementary perspectives of the same cell, providing a more complete and holistic view of cellular processes [22, 33]. For example, scRNA-seq reveals gene expression levels, scATAC-seq provides insights into chromatin accessibility that regulates gene expression, and CITE-seq captures protein abundance on the cell surface. By integrating these multiple data layers, we can uncover a deeper understanding of cell differentiation, disease mechanisms, and therapeutic responses [8]. However, the challenge lies in effectively combining these different data types to maximize their biological insights, especially considering the heterogeneity, sparsity, and scale of single-cell data [31].

In recent years, several computational methods have been developed to tackle the problem of multi-modal data integration. Statistical techniques, such as Canonical Correlation Analysis (CCA) and Multiple Correspondence Analysis (MCA), align features from different modalities by finding linear relationships between them [25]. For example, Seurat's Weighted Nearest Neighbor (WNN) analysis incorporates CCA to balance shared and modality-specific signals, effectively integrating RNA-seq and protein data [17]. Matrix factorization methods, such as MOFA+ (Multi-Omics Factor Analysis), decompose data matrices into shared and unique components to uncover underlying factors driving multi-modal variation [12]. Although these methods have successfully integrated multi-modal data, they are limited by their inability to capture complex non-linear relationships, which are prevalent in single-cell data integration.

Machine learning-based approaches address these limitations by leveraging non-linear transformations to map data into shared latent spaces [15]. For instance, Deep Canonical Correlation Analysis (DeepCCA) [25] extends CCA by using deep neural networks

to learn more complex relationships between modalities [4]. Autoencoders, such as those implemented in SIMBA and scVI [18], compress high-dimensional data into lower-dimensional embeddings, enabling efficient integration and clustering [13]. Generative models like totalVI [2] further improve integration by probabilistically reconstructing missing data and accounting for technical noise. However, these methods can suffer from computational inefficiency, particularly with large-scale datasets, and are often prone to overfitting in cases of sparsity and batch effects [18]. Recent frameworks, such as scJoint and MultiVI, have attempted to address these issues by integrating batch correction and feature alignment directly into their architectures.

To address these issues, we propose scSNFGNN, a novel integration method that combines Similarity Network Fusion (SNF) [11] with Graph Neural Networks (GNNs) [1]. scSNFGNN leverages the iterative refinement of similarity networks through SNF, capturing relationships across multiple modalities and reducing batch effects [3]. After fusion, the integrated data is processed using GNNs [16], which use graph convolutional layers to learn robust [14], low-dimensional embeddings that preserve both local and global structures [5]. GNNs are well-suited to handle complex, non-linear relationships, making scSNFGNN particularly effective for multi-modal data where interactions across molecular layers are intricate and dynamic [24]. The use of GNNs also ensures that the integrated data can scale efficiently [24], addressing computational concerns when processing large datasets. This hybrid approach overcomes key limitations of existing methods, improving clustering accuracy, robustness against noise and sparsity, and better preservation of modality-specific information [23].

The novelty of scSNFGNN lies in its ability to combine SNF's iterative fusion of data with the powerful learning capabilities of GNNs. Graph-based methods, such as Similarity Network Fusion (SNF), iteratively combine similarity matrices from different data sources to capture both local and global relationships between cells [7]. This integration allows for more accurate, biologically meaningful clustering of cells, preserving cellular heterogeneity while mitigating the effects of noise and batch differences. In addition to single-cell data, this technique has shown success in other domains, such as drug discovery, where it is used to integrate diverse biological data types to predict drug-target interactions [32]. This success outside of single-cell research suggests that scSNFGNN could have broad applicability in other fields that require multi-modal data integration [29].

This paper is organized as follows: Section 2 reviews the challenges and current computational methods for multi-modal data integration, highlighting their limitations. Section 3 presents scSNFGNN, describing its algorithm and how the integration of SNF with GNNs addresses the challenges identified in previous methods [28]. Section 4 evaluates scSNFGNN through experimental results, demonstrating significant performance improvements, including ARI and AUC, showcasing its superiority in handling multi-modal data [30, 26]. Finally, Section 5 discusses the implications of scSNFGNN for future single-cell research and its potential applications in other areas of data integration.

2 Methods

2.1 Datasets

This section uses the 10XPBMCs multimodal dataset, which can be downloaded from <https://support.10xgenomics.com>, where the basic information of this dataset is shown in table 1.

This paper compares the traditional spectral clustering model [27], the SIMBA model [6], the scMIC model [35], the scMIC+SNF method, and two other state-of-the-art multimodal integration methods: the Guanlab-Dengkw method [19] and the DCCA method [34].

2.2 Single-cell data preprocessing

For scRNA-seq data, genes with low expression in cells are filtered, and raw counts are normalized and log-transformed to help reduce technical noise and batch effects between different cells [33]. Variable gene selection is then performed to retain genes that are more differentially expressed in different cells in the single-cell RNA sequencing data, and which are considered "variable" and therefore more likely to contain important information about cell state and type.

For scATAC-seq data, peaks appearing in very few cells were filtered. Then select variable peaks based on SVD: SVD decomposition is performed on the data, the first k principal components are identified and selected using the inflection plot of the variance ratio, the peaks with significant effects are automatically selected by calculating the loadings and using the knee detection algorithm, and finally the peaks selected by each principal component are combined and labeled as "variable peaks". This process helps to filter out non-informative peaks, improving the quality and accuracy of the data and speeding up the training process [9].

Figure 1 illustrates a two-stage pipeline for integrating scRNA-seq and scATAC-seq data at single-cell resolution. In panel a, the workflow begins with the independent preprocessing of gene-expression and chromatin-accessibility matrices, yielding modality-specific cell-cell similarity maps whose darker shades mark stronger affinities. Each similarity map is then converted into a weighted k -nn graph, coloured pink for RNA and blue for ATAC, so that every node represents a cell and edge thickness reflects the magnitude of similarity within that modality. An iterative fusion algorithm repeatedly transfers edge weights between these graphs, gradually aligning their local topologies; by the final iteration, edges confirmed by both modalities are reinforced, producing a single fused network (black edges with yellow highlights) that encapsulates complementary transcriptional and epigenetic relationships among all cells. Panel b shows how this fused network is turned into a low-dimensional embedding through self-supervised learning: a random subset of true edges is masked, and a contrastive objective drives the model to reconstruct them, forcing the embedding to capture higher-order structure while suppressing noise. The resulting latent vectors regenerate a denoised similarity matrix and, when projected with UMAP or t-SNE, reveal well-separated clusters corresponding to distinct cell types (six are coloured here). Thus, the figure conveys an end-to-end strategy that first harmonises multi-modal graphs and then learns modality-agnostic representations, enabling accurate clustering, intuitive visualisation, and

Table 1: Clustering results of seven single-cell multimodal integration methods

Methods	Spectral clustering	SIMBA	scSNFGNN	scMIC	SNF+scMIC	Guanlab-dengkw	DCCA
ARI	0.0991	0.4854	0.5584	0.1142	0.3523	0.2683	0.3375
NMI	0.3633	0.6839	0.7274	0.2841	0.5732	0.5164	0.5798
AMI	0.3608	0.6822	0.7260	0.2802	0.5710	0.5140	0.5777

downstream biological discovery from heterogeneous single-cell datasets.

2.3 Graph embedding and training

In the multi-omics integration process, graphs for scRNA-seq and scATAC-seq were constructed following the respective single-cell RNA-seq and ATAC-seq analysis steps. Gene activity scores were calculated for scATAC-seq data by considering peaks around the Transcription Start Site (TSS). These scores were based on peak weights determined by their proximity to the TSS. The gene scores were then normalized to gene size. Once gene scores were obtained, a similar procedure to that described in 'Similarity Network Fusion' was used to infer edges between cells profiled by scRNA-seq and scATAC-seq.

We utilized a directed graph $G_2 = (V_2, E_2)$ as input, where V_2 represents a set of entities (vertices) and E_2 represents a set of edges. Each edge $e_2 = (u_2, v_2)$ connects a source entity u_2 to a destination entity v_2 . It was assumed that each entity had a distinct known type, such as cells or peaks. Graph embedding techniques were employed to learn a D -dimensional embedding vector for each $v_2 \in V_2$ by optimizing a link prediction objective using stochastic gradient descent, with $D = 100$ chosen for our experiments. The complete embedding matrix was denoted as $\Theta \in \mathbb{R}^{|V_2| \times D}$, and the embedding for an entity v_2 was represented as θ_{v_2} . For an edge $e_2 = (u_2, v_2)$, the score for e_2 was denoted as $s_{e_2} = \theta_{u_2} \cdot \theta_{v_2}$, and a multi-class log loss was optimized.

$$\mathcal{L}_{e_2} = -\log \frac{\exp(s_{e_2})}{\sum_{e_2' \in N_e} \exp(s_{e_2'})} w_{e_2} \quad (1)$$

Here, N_e represents a set of 'negative sampled' candidate edges generated by corrupting e_2 , and w_{e_2} denotes the edge weight. By default, the edge weight is the relation weight but can differ for each edge within a specific relation type. The objective of this log loss function is to enhance the score for all $(u_2, v_2) \in E$ while minimizing it for $(u_2, v_2) \notin E$.

We used a PyTorch-BigGraph-based graph model for training, generating initial embeddings for all entities (nodes). However, entities of varying types (such as single-cell multimodal data) exhibit distinct edge distributions and may reside on separate manifolds within the latent space. To ensure the comparability of embeddings for entities of different types, we applied the Softmax function to transform the feature embeddings based on the first-order similarity between cells (reference) and features (query).

2.4 Indicators

To evaluate the performance of various methods, this paper uses three widely used evaluation metrics including Adjusted Rand Index (ARI) [21], Normalized Mutual Information (NMI) [10], and

Adjusted Mutual Information (AMI) [20]. ARI is used to measure the similarity between the model's predicted categories and the true categories, defined as:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}} \quad (2)$$

The a_i and b_i denote the number of cells in the i th real cluster and the i th predicted cluster, respectively, and n_{ij} denotes the number of shared cells in the i th real cluster and the j th predicted cluster. NMI is used to assess the consistency between predicted and true categories. Let $T = t_1, t_2, \dots, t_n$ and $Y = y_1, y_2, \dots, y_n$ denote the true and predicted categories of cells, respectively, then define:

$$\text{NMI}(T, Y) = \frac{2\text{MI}(T, Y)}{H(T) + H(Y)} \quad (3)$$

The $\text{MI}(\cdot)$ and $H(\cdot)$ denote information entropy and mutual information, respectively. Considering that MI is higher if two clusters are of larger magnitude, regardless of whether there is more information to share, AMI adjusts the mutual information score, defined as:

$$\text{AMI}(T, Y) = \frac{\text{MI}(T, Y) - E(\text{MI}(T, Y))}{[H(T) + H(Y)]/2 - E(\text{MI}(T, Y))} \quad (4)$$

The $E(\cdot)$ denotes the mathematical expectation. The values of ARI and AMI are in the range of $[-1, 1]$, and the values of NMI are in the range of $[0, 1]$. The larger the values of these metrics, the better the performance of clustering.

3 Results

3.1 Similarity network fusion module

The similarity network fusion (SNF) approach constructs a network for each modal data, and then based on the principle of "neighbor information sharing", the constructed different network structures are efficiently fused into a full-spectrum network representing the underlying data. The fused network is capable of capturing the shared and complementary information from various data sources. At the same time, SNF integrates multiple modal data based on similarity, first calculates the similarity between each pair of cells, and does not show the batch they belong to, which means that even if there are similarities between cells from different batches, they are still captured and retained. Thus similarity information from different batches is also considered, which mitigates the impact of the batch effect.

3.1.1 Similarity-based data composition for each modality.

For the network of individual modalities to be constructed $G = (V, E)$, vertices V correspond to cells $\{x_1, x_2, \dots, x_n\}$, and edges E are weighted by the degree of similarity between cells. The edge weights are represented by the similarity matrix $W \in \mathbb{R}^{n \times n}$ denoted

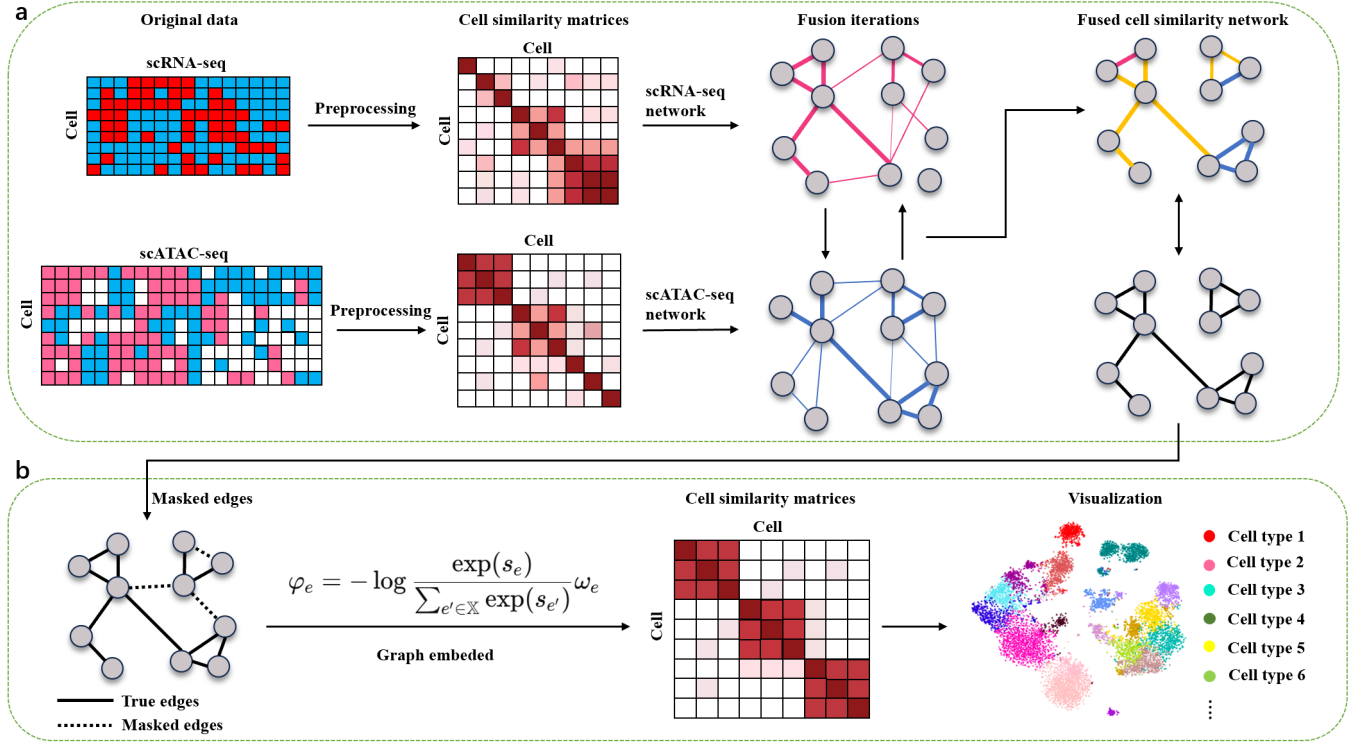


Figure 1: The scSNFGNN model framework, (a) represents the process of graph construction based on cell similarity of scRNA-seq, including the data preprocessing process and the construction of similarity matrix, (b) represents the process of composition by features, (c) represents the iterative process of graph fusion based on cell similarity. (d) represents the integration of multiple composition methods to obtain the final fusion matrix, (e) represents the cell feature matrix after depth map embedding, (f) represents the visualization results of clustering, and (g) represents the similar process of scATAC-seq data processing as (a).

by $W(i, j)$ denotes the cell x_i and cell x_j similarity between cells and the edge weights are then determined using a scaled exponential similarity kernel:

$$W(i, j) = e^{-\frac{\rho^2(x_i, x_j)}{\mu \varepsilon_{i,j}}} \quad (5)$$

$\rho(x_i, x_j)$ is the Euclidean distance between cells, the μ is a hyperparameter that can be set empirically (typically in the range $[0.3, 0.8]$), that $\varepsilon_{i,j}$ is used to eliminate the scaling problem, defined here,

$$\varepsilon_{i,j} = \frac{\frac{1}{n} \sum_{i=1}^n (\rho(x_i, N_i)) + \frac{1}{n} \sum_{j=1}^n (\rho(x_j, N_j)) + \rho(x_i, x_j)}{3} \quad (6)$$

$\rho(x_i, N_i)$ is the x_i the average of the distances to each of its neighboring cells. This distance metric applies to continuous variables and can be used to measure discrete variables using chi-square distances and to deal with binary variables using a consistency metric.

To compute the fusion matrix from multiple histological networks, a fully sparse kernel is defined on the vertex set V . $P = D^{-1}W$, where P is a normalized weight moment, D is a diagonal matrix, $D(i, i) = \sum_j W(i, j)$, so that $\sum_j P(i, j) = 1$. However, this normalization may lead to numerical instability due to the self-similarity involved on the W diagonal and a more optimal

normalization process can be used:

$$P(i, j) = \frac{W(i, j)}{2 \sum_{k \in V, k \neq i} W(i, k)}, \quad j \neq i \quad (7)$$

When $j = i$, $P(i, j) = 0.5$. Let N_i denote the set of "neighbors" in G containing x_i the set of "neighbors" of G . Given a graph G , measure the local affinity by k -nearest neighbors (KNN):

$$S(i, j) = \frac{W(i, j)}{\sum_{k \in N_i} W(i, k)}, \quad j \in N_i \quad (8)$$

When $j = i$, $S(i, j) = 0$. This operation sets the similarity between disjoint points to 0.

In this paper, we will take the above method as a benchmark to compose the data of multiple modalities. Then there are some differences in data similarity and structural patterns in different ways of composition. We will follow up to explore the effects of other ways of composition on the algorithm results at the same time:

- Cosine similarity was used instead of Euclidean distance: In constructing the network using similarity, changing Eq. (1) in the $\rho(x_i, x_j)$ definition, cosine similarity is used instead of Euclidean distance. Euclidean distance takes into account the difference of cells in each feature dimension, and therefore, cells are less similar when they have large

differences in multiple features. In contrast, cosine similarity calculates the cosine of the angle between two cells based on their orientation in the feature space without considering their sizes and is suitable for similarity comparisons of high-dimensional data, independent of the size and scale of the vectors.

- **Neighborhood Definition of KNN Using Cardinality Distances** N_i : Cell-gene expression matrix by scRNA-seq $A \in R^{n \times m}$. As an example, the gene data are discretized using the split-box technique, and then the chi-square distance between cells is obtained:

$$\chi^2(a_i, a_j) = \sum_t \frac{(a_{it} - a_{jt})^2}{a_{it} + a_{jt}} \quad (9)$$

Of these, the a_{it} and a_{jt} denote the values taken by the two cells after discretization on the t th gene, respectively, and the nearest K neighbors are found to replace the neighbor set N obtained by the original method according to the calculation.

3.1.2 Similarity Network Fusion.

For the modal network described above, P contains all the information about the similarity of each cell to all the remaining cells, while S encodes only the similarity of each cell to the k most similar cells. In the similarity fusion process, P is always the initial state and S is the kernel matrix, which can efficiently capture the local structural capacity of the graph as well as improve the computational efficiency.

Assuming that there are 2 histologies, the data can be constructed as in the previous steps to obtain $W^{(v)}$, $P^{(v)}$ and $S^{(v)}$. $v = 1, 2$. The key step of SNF is to iteratively update the similarity matrix corresponding to each data:

$$P_{t+1}^{(1)} = S^{(1)} \times P_t^{(2)} \times (S^{(1)})^T \quad (10)$$

$$P_{t+1}^{(2)} = S^{(2)} \times P_t^{(1)} \times (S^{(2)})^T \quad (11)$$

where $P_{t+1}^{(1)}$ is the state matrix of the first histology data after t iterations and $P_{t+1}^{(2)}$ is the state matrix of the second histological data, the process updates the state matrix each time thus generating two parallel interchangeable diffusion processes that update the overall state matrix after t iterations:

$$P^{(c)} = \frac{P_t^{(1)} + P_t^{(2)}}{2} \quad (12)$$

To reduce the noise, update the rule again to:

$$P_{t+1}^{(1)}(i, j) = \sum_{k \in N_i} \sum_{l \in N_j} S^{(1)}(i, k) \times S^{(1)}(j, l) \times P_t^{(2)}(k, l) \quad (13)$$

For $P_{t+1}^{(1)}$, it can be observed that if x_i and x_j have common "neighbor" nodes in both similarity matrices, then it is more likely that they belong to the same cluster, and if they are not too similar, their similarity in the first histological data can be represented by the second histological data, thus indicating that this type of similarity information can be propagated through the fusion process. Finally, expanding to m histological data ($m > 2$), the state matrix update

rule is as follows:

$$P^{(v)} = S^{(v)} \times \left(\frac{\sum_{k \neq v} P^{(k)}}{m-1} \right) \times (S^{(v)})^T, v = 1, 2, \dots, m \quad (14)$$

In such a fusion approach, low-weight edges representing weakly similar cellular connections disappear, which can reduce noise to some extent, while strong similarities (high-weight edges) present in one or more networks are added to other networks, and this nonlinear property enables SNF to fully exploit the local structure of the network and to integrate the shared and complementary information across networks.

3.2 Deep graph learning module

Regarding the heterogeneous graph construction for multimodal integration, here the cell-feature and cell-cell relationships are mapped to the graph structure, our method is used to infer cell boundaries by calculating the weighted sum of upstream and downstream peaks of the TSS at the transcription start site as the gene score of each gene, and the cell boundaries are inferred by the gene activity scores, and in this dissertation, the relationship of cell edges is inferred by the original method has been improved by integrating the information of several modalities through the SNF method.

3.2.1 Cell-gene relationships.

In constructing the graph of cells and genes, if a gene is expressed in a given cell, an edge is added between the cell and the gene, and the weight of each edge takes a split-box process that assigns and encodes different levels of gene expression values, while preserving the original distribution, as follows: The distribution of non-zero values in the gene expression matrix is first approximated and normalized based on k-mean clustering, and then the continuous non-zero values are divided into n intervals with widths depending on the one-dimensional k-mean clustering, where the values of each interval are assigned to the same cluster centers, at which point the continuous matrix is converted to a discrete matrix, denoted by $1, \dots, n$ denotes the expression level of n genes. The graph is then constructed by weighting relationships with cells and genes as nodes and n different gene expression levels as edges.

3.2.2 Cell-peak relationships.

The peak-cell matrix is first binarized, with "1" indicating the presence of at least one read within a peak, and "0" otherwise. When DNA sequence features are available, this graph can be further enriched by encoding k-mer and motif sequence entities as nodes, and using the presence of these entities within peaks as edges between these additional nodes and the peak node, where a weight of 0.02 is assigned to the relationship between the k-mer and the peak and a weight of 0.2 is assigned to the relationship between the motif to construct the original peak-cell graph's Relationships.

3.2.3 Inferring cell-cell relationships based on SNF methods.

After constructing similarity networks for scRNA-seq and scATAC-seq data respectively, the two similarity networks were fused based on the SNF method, and after information fusion and updating to the convergence of network iterations, at this time, the weak inter-cell similarity disappeared, and the strong similarity was added or the weight was strengthened, and the similarity fusion network

with comprehensive information was obtained. And the intercellular connections of the heterogeneous map are determined based on the connections between similar fusion networks. Compared with the traditional method, which infers cell boundaries by calculating the gene activity scores, the SNF method can more comprehensively take into account the inter-cell similarities and differences, and at the same time, it can counteract the influence of batch effect to some extent.

3.2.4 Single-cell multimodal data integration process.

Based on the constructed multi-relational graph of biological entities $G = (V, E)$, where V is the set of vertices, E is the set of edges, and the edge $e = (u, v)$ between the source entity u and the target entity v , a graph embedding method is used to predict the target for each entity by optimizing the links $v \in V$. Learning a D -dimensional embedding vector, the whole embedding matrix is denoted as $\theta \in R^{|V| \times D}$, where the embedding of entity v is denoted as θ_v , to learn the edge corresponding to entity $e = (u, v)$, define the fraction of $es_e = \theta_u \cdot \theta_v$, and then for each edge e set a multi-category logarithmic loss function:

$$\varphi_e = -\log \frac{\exp(s_e)}{\sum_{e' \in \mathbf{N}} \exp(s_{e'})} \omega_e \quad (15)$$

where $\mathbf{N}$ is the set of "negatively sampled" candidate edges generated by destroying e , and ω_e is the weights of the edges, and the purpose of this function is to maximize the scores of all the $(u, v) \in E$ scores and minimize $(u, v) \notin E$. This function aims to maximize the scores of all edges and minimize the scores of all edges, i.e., the model is expected to correctly predict edges that are known to be present and increase their scores as much as possible while minimizing the misprediction of edges that are not present and decrease their scores. Negative sampling, on the other hand, generates multiple negative samples by destroying edges using sources or targets that are uniformly sampled in nodes with the correct type, and some function of the degree of such nodes proportionally samples negative samples to produce more informative embeddings. However, this negative sampling may produce overfitting, so regularization is introduced to control it:

$$\varphi_{reg} = \sum_{t=1}^{N_e} \varphi_t + \lambda \sum_{u \in N} \sum_{d=1}^D \theta_{ud}^2 \quad (16)$$

φ_t is the logarithmic loss function of the category of the t th edge and N_e is the total number of edges, and λ denotes the weight decay parameter, set $\lambda = \frac{C}{N_e}$, C is a constant, and finally the embedding graph of cells is trained to realize the integration of single-cell multimodality.

3.3 Analysis of clustering results

The results from the comparison of single-cell multimodal integration methods underscore the superior performance of GNN-based approaches (Table 1), particularly scSNFGNN, which achieved the highest scores across all metrics (ARI, NMI, and AMI). This highlights the efficacy of GNNs in capturing both local and global structures within complex, high-dimensional multimodal data, making them exceptionally well-suited for integrating and clustering single-cell datasets.

Table 2: 10XPBMCs Dataset

Dataset	Cells	Genes (scRNA)	Peaks (scATAC)	Cell types
PBMC-10k	11909	36601	108377	19
Chen-2019	9190	28930	241757	22

Table 3: Transformation of clustering metrics after incorporating SNF methods

Methods	ARI	NMI	AMI
SNF+SIMBA	+15.04%	+6.36%	+6.42%
SNF+scMIC	+208.49%	+101.76%	+103.78%

SIMBA also demonstrated strong performance, significantly outperforming traditional spectral clustering. This suggests that while SIMBA effectively handles the integration of multimodal data, it still lacks the advanced modeling capabilities of GNNs, which limits its performance relative to scSNFGNN.

DCCA and Guanlab-Dengkw, while performing better than scMIC and SNF+scMIC, did not reach the high performance of scSNFGNN and SIMBA. These methods effectively integrate multimodal data but are limited by their inability to fully capture the intricate relationships within the data, which GNNs handle more proficiently.

The integration of SNF with scMIC showed noticeable improvements over scMIC alone, indicating that SNF can mitigate some of scMIC's limitations by better integrating multimodal data and reducing batch effects. However, SNF+scMIC still lagged behind more advanced methods like SIMBA, DCCA, and especially scSNFGNN.

Overall, the results indicate that while traditional and non-GNN-based methods can provide reasonable performance, the advanced capabilities of GNNs in modeling complex interactions make them the most effective approach for single-cell multimodal data integration and clustering. This suggests a pivotal role for GNNs in future developments and applications in this field.

For Table 3, Using the ARI as a benchmark to analyze the clustering effect of each method on the 10xPBMCs dataset, it can be found that the scSNFGNN method has the best ARI, followed by the DCCA method, the Guanlab-dengkw method, and lastly, the GNN and SNF methods, so the performance of the GNN-based and SNF methods alone does not seem to be that good. However, the clustering results obtained after improving the similarity fusion matrix obtained from SNF in conjunction with the HGNN and GNN methods, the ARI improvement of the improved GNN by 208.49% (0.1142→0.3523) far exceeds the ARI improvement effect of the improved HGNN (+15.04%).

To visualize the clustering results more intuitively and to validate the effectiveness of the improved method in performing single-cell multimodal integration, this paper uses t-SNE to visualize potential representations of the fusion module data or outputs. The left and right plots are colored by the model-predicted and true category labels, respectively, and it can be observed that the heights of the cell type distributions are kept consistent.

The single-cell multimodal data were integrated by the scSNFGNN method, the predicted clusters of different cell categories achieved

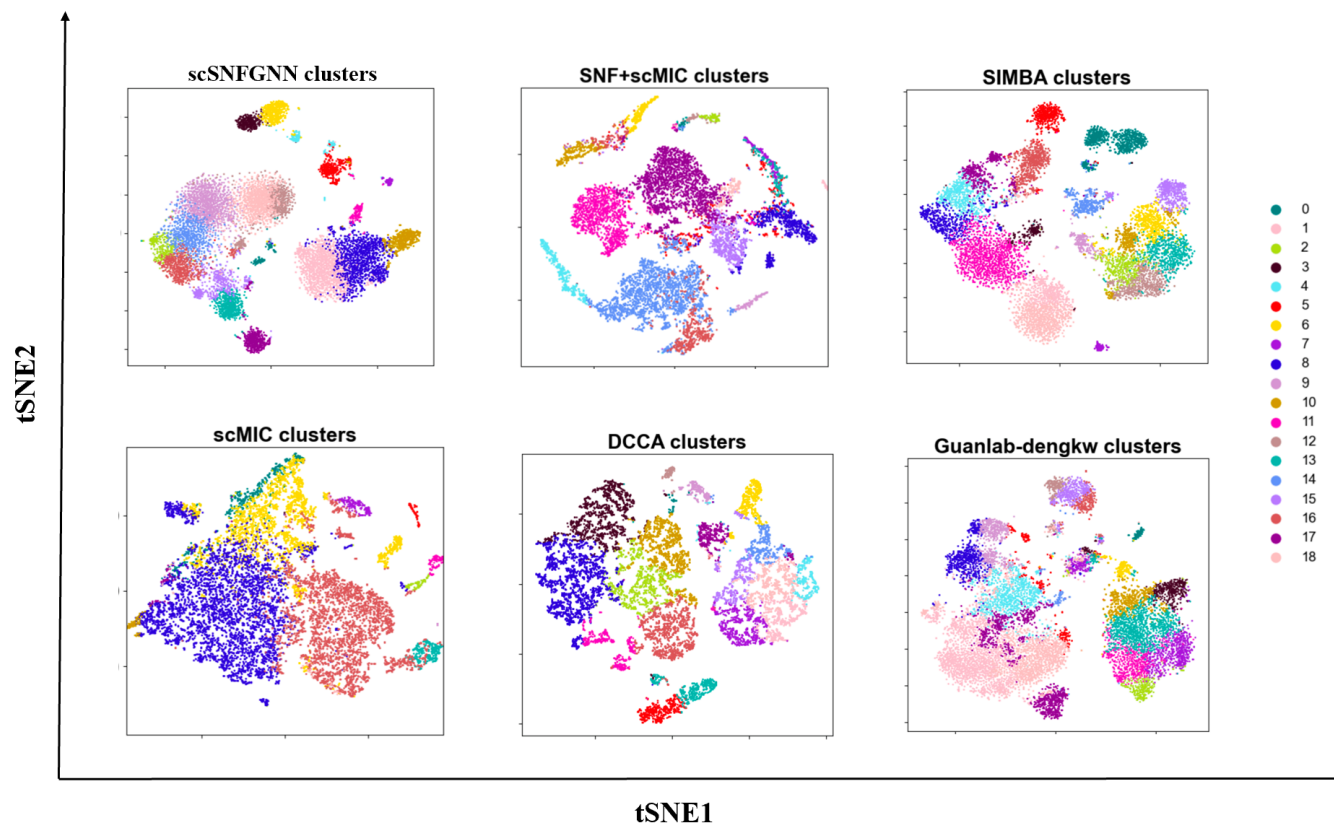


Figure 2: The figure represents the different contrast models of single-cell modal data clustering visualization, indicating the results of multimodal integration.

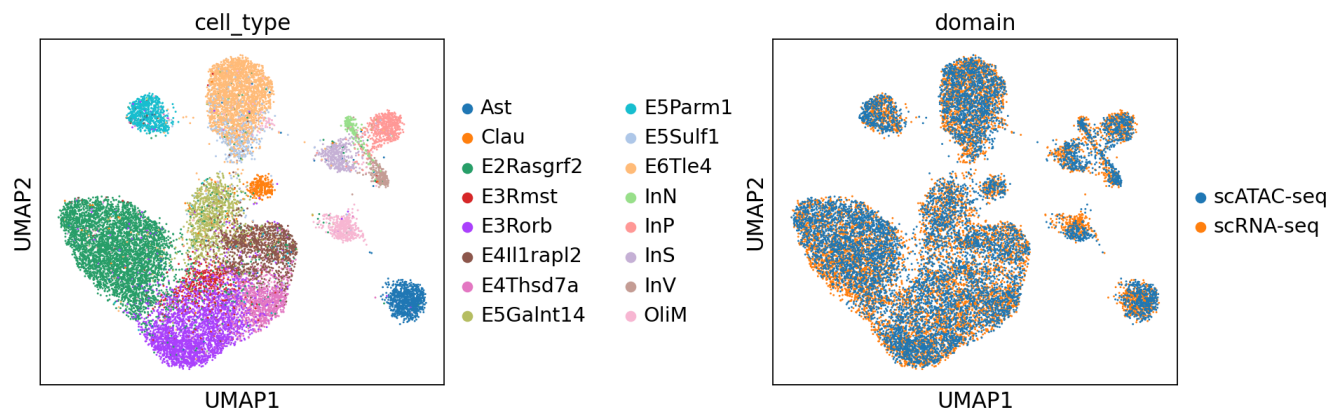


Figure 3: Joint UMAP Embedding of scRNA-seq and scATAC-seq from the Chen-2019 Dataset.

good separation and the boundaries between the clusters were clearly defined, implying that the method was very effective for single-cell multimodal integration. Second, observing the t-SNE plots of the scMIC method before and after the improvement, it can be found that the model after integrating the SNF method improves the clustering effect of cells.

In this study, clustering is used to compare and analyze the performance of established modeling methods and the improvement effect of SNF. In the first stage, the dataset and evaluation indexes were determined. The remaining two popular single-cell multi-omics data integration methods were introduced. The integrated multimodal data was then clustered, and the clustering index results

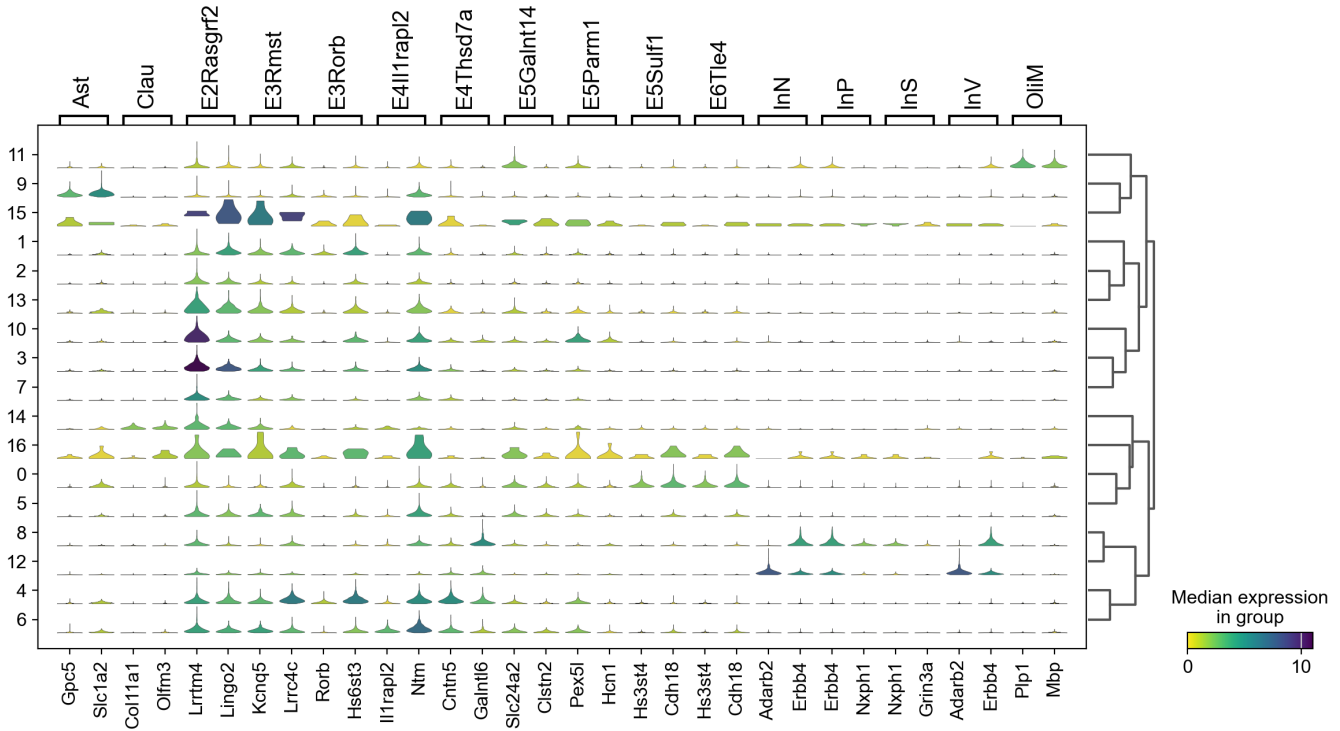


Figure 4: Layer and cell-type-specific marker-gene landscape revealed by a stacked violin plot

Table 4: Effects of different compositional approaches on the clustering results of two models

Methodologies	Composition	ARI	NMI	AMI
scSNFGNN	cosine distance	0.5719	0.7582	0.7351
	chi-square distance	0.5443	0.7179	0.7206
SNF+scMIC	cosine distance	0.3811	0.6047	0.5928
	chi-square distance	0.3024	0.5223	0.5176

were compared with the reduced dimensionality graphs of t-SNE. It was found that the best-performing model was the joint method of SNF+HGNN. Moreover, the similarity fusion network based on SNF showed a significant improvement in the GNN model. The second similar fusion network based on SNF also demonstrated significant improvement in the GNN model. Figure 3 illustrates the joint embedding of scRNA-seq and scATAC-seq data from the Chen-2019 dataset using a multimodal integration algorithm. In the left panel, cells are colored by cell type, revealing 18 distinct clusters that correspond to known biological cell identities. In the right panel, cells are colored by modality, with blue representing scATAC-seq and orange representing scRNA-seq. The strong overlap between modalities within each cluster indicates effective alignment and integration across the two data types. This demonstrates that the algorithm successfully removes modality-specific batch effects and preserves cell-type-specific structure in the shared latent space.

Figure 3 presents a stacked-violin matrix of marker gene expression across 17 single-cell clusters. Violin width shows expression spread, and a yellow-to-purple gradient marks median expression levels. Top labels (e.g., Ast, E3Rorb, OliM) denote inferred cell types, while the side dendrogram orders clusters by transcriptional similarity, linking gene enrichment to cell-type relationships in one compact view.

3.4 Parameter sensitivity analysis

In scSNFGNN composition, the number of nearest neighbors k , controls the sparsity of the constructed cell graph, which may have an effect during similarity fusion. To examine the effect of k on the clustering performance, in this paper, we vary the value of k in the $\{30, 40, 50, 60\}$ range of varying the value of k and still perform experiments on the 10xPBMCs dataset presented in Chapter III. As shown in Fig. 7 shows, both models achieve the best clustering results when the value of k is set to 60, and the clustering effect is reduced when the value of k is either small or large, indicating that scSNFGNN composition needs to select the appropriate k neighborhood parameter.

3.5 Effect of SNF composition on clustering accuracy

scSNFGNN is based on distance for similarity composition, here different distances have some differences in the structural patterns of data similarity and graphs, where Euclidean distance takes into account the difference in the size of the feature space and cosine

similarity focuses on the direction of the feature space. Then the KNN method is used to obtain k nearest neighbor cells based on the similarity calculated earlier, and since the cell boundaries measured by similarity are relatively fuzzy and the distribution may be inconsistent, the split-box technique is used to calculate the cardinal distances of the cells in place of the set of neighbors, N , obtained by directly using the similarity.

To investigate the effect of these two compositional approaches on the clustering effect of the final two joint models, the experiment is still conducted on the 10xPBMCs dataset, where the number of nearest neighbors is fixed $k = 50$, the results of the two models obtained under the compositional approaches of using cosine similarity instead of Euclidean distance or using chi-square distance instead of similarity calculation are as follows:

Based on the results, it was observed that using cosine distance instead of Euclidean distance can improve the clustering effect due to the sparseness of the gene expression matrix or cell peak matrix. On the other hand, using chi-square distance instead of similarity calculation does not significantly affect the scSNFGNN model, but it does reduce the clustering effect of the improved scMIC joint model.

In this study, a parameter sensitivity analysis is conducted for the established joint model to explore the influence of the compositional approach of the similarity fusion method on the model training and clustering effect. The multimodal integration model is realized by changing the k -nearest neighbor parameter of KNN composition, using cosine distance instead of Euclidean distance to calculate similarity, and adopting the split-box technique and chi-square distance instead of similarity calculation. It is finally found that the appropriate k -nearest neighbor parameter, cosine distance instead of Euclidean distance can improve the clustering effect of the model to some extent, and the model is also verified to have better stability and robustness.

4 Discussion and conclusion

This study explores the use of single-cell multimodal data to improve cell clustering by combining a novel SNF method inspired by heterogeneous graph neural networks. By combining SNF with established graph neural network techniques, the study aims to effectively combine different types of data. SNF helps to create detailed structural information, allowing for better connections between cells, which is important for training graph neural networks. Using the 10xPBMCs dataset, the combined model shows significant improvements in clustering compared to previous methods. The SNF-based fusion network enhances the discriminative potential and clustering quality of representations, leading to better performance.

Additionally, the study analyzes the impact of model parameters, such as the K -neighborhood parameter and the compositional approach of scSNFGNN, on clustering results. Looking ahead, the research emphasizes the need to test the model's effectiveness across different datasets to ensure its general applicability. Future work aims to further investigate the fusion patterns of scSNFGNN in different model structures and explore its compatibility with various deep learning methodologies to improve clustering in single-cell multimodal analysis.

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